

HDOC 26

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with options 1 to 4 to the user.
3. Read the user's choice.
4. Use a switchstatement to call the respective function or exit if choice is 4.
5. Function **rightAlignedIncreasingStar()**:
 - Read n. Validate $n > 0$. Loop ifrom 1 to n.
 - Inside, loop jfrom 1 to n - ito print spaces (" ").
 - Loop kfrom 1 to ito print asterisks ("* "). After inner loops, print newline.
6. Function **rightAlignedIncreasingNumber()**:
 - Read n. Validate $n > 0$. Loop ifrom 1 to n.
 - Inside, loop jfrom 1 to n - ito print spaces (" ").
 - Loop kfrom 1 to ito print numbers (k). After inner loops, print newline.
7. Function **rightAlignedDecreasingStar()**:
 - Read n. Validate $n > 0$. Loop ifrom ndown to 1.
 - Inside, loop jfrom 1 to n - ito print spaces (" ").
 - Loop kfrom 1 to ito print asterisks ("* "). After inner loops, print newline.
8. Repeat steps 2-4 until choice == 4.
9. Stop

2. Test-Case Table

TC ID	Scenario Description	Inputs	Expected Output
TC-01	Right-aligned increasing star (Valid)	choice: 1, n: 4	Row 1: 3 spaces + 1 star, Row 4: 0 spaces + 4 stars
TC-02	Right-aligned increasing numbers (Valid)	choice: 2, n: 4	Row 1: 3 spaces + "1", Row 4: "1 2 3 4"
TC-03	Right-aligned decreasing star (Valid)	choice: 3, n: 4	Row 1: 0 spaces + 4 stars, Row 4: 3 spaces + 1 star
TC-04	Invalid dimension (Boundary)	choice: 2, n: 0	Error message: Positive integer required.
TC-05	Invalid menu choice	choice: 6	Error message: Invalid choice. Please select 1-4.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes
void rightAlignedIncreasingStar(void); void
rightAlignedIncreasingNumber(void); void
rightAlignedDecreasingStar(void);

int main() {
    int choice; do {
        printf("\n--- Right-Aligned Patterns Menu ---\n"); printf("1. Right-aligned
        increasing triangle\n"); printf("2. Right-aligned increasing number triangle\n");
        printf("3. Right-aligned decreasing triangle\n"); printf("4. Exit\n");
        printf("Enter your choice: "); scanf("%d",
        &choice);

        switch (choice) {
            case 1: rightAlignedIncreasingStar(); break; case 2:
            rightAlignedIncreasingNumber(); break; case 3:
            rightAlignedDecreasingStar(); break; case 4: printf("Exiting
            program...\n"); break;
            default: printf("Error: Invalid choice. Please enter 1-4.\n");
        }
    } while (choice != 4);

    return 0;
}

void rightAlignedIncreasingStar(void) { int n, i, j, k;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }

    for (i = 1; i <= n; i++) {
        // Print leading spaces
        for (j = 1; j <= n - i; j++) { printf(" ");
        }
        // Print asterisks
        for (k = 1; k <= i; k++) { printf("* ");
        }
        printf("\n");
    }
}

void rightAlignedIncreasingNumber(void) { int n, i, j, k;
    printf("Enter number of rows n: "); scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }
```

```

    for (i = 1; i <= n; i++) {
        // Print leading spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // Print numbers
        for (k = 1; k <= i; k++) {
            printf("%d ", k);
        }
        printf("\n");
    }
}

void rightAlignedDecreasingStar(void) {
    int n, i, j, k;
    printf("Enter number of rows n: ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Positive integer required.\n"); return; }

    for (i = n; i >= 1; i--) {
        // Print leading spaces
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // Print asterisks
        for (k = 1; k <= i; k++) {
            printf("* ");
        }
        printf("\n");
    }
}

```